Erdős Problem #1209

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Abstract

We give complete solutions to the following three problems.

(1) Let $A = \{a_1 < a_2 < \cdots\}$ be a sequence of integers tending to ∞ sufficiently fast. If there exists an integer n such that every $n + a_k$ is prime, must there be infinitely many such n?

(2) Under the same hypotheses, what happens if "prime" is replaced by "squarefree"?

(3) Does there exist an integer n such that $n + 2^{2^k}$ is prime for every $k \geq 0$?

For (1) and (2) we prove a stronger statement: for every prescribed growth rate one can construct a sequence for which there is exactly one good shift. For (3) we prove that no such integer n exists.

1 Introduction and notation

Throughout, a *shift* means an integer $n \in \mathbb{Z}$. A nonzero integer m is *squarefree* if no square $d^2 > 1$ divides m; equivalently, $|m|$ is not divisible by the square of any prime. For a nonzero integer m, we write $v_2(m)$ for the largest integer $r \geq 0$ such that $2^r \mid m$.

The phrase "tending to ∞ sufficiently fast" is made precise by the following standard strengthening: given any function $g\colon \mathbb{N} \to \mathbb{N}$ with $g(k) \to \infty$, we construct a sequence $A = \{a_k\}$ satisfying $a_k > g(k)$ for every k. This is more than enough for the original formulation.

2 A diagonal construction for Questions (1) and (2)

The key point is that the prime and squarefree questions can be killed simultaneously.

Theorem 2.1. *Let $g\colon \mathbb{N} \to \mathbb{N}$ be any function with $g(k) \to \infty$. Then there exists a strictly increasing sequence of primes*

$$B = \{b_1 < b_2 < \cdots\}$$

such that:

(i) *$b_k > g(k)$ for every k;*

(ii) *for an integer n, all of the numbers $n + b_k$ are prime if and only if $n = 0$;*

(iii) *for an integer n, all of the numbers $n + b_k$ are squarefree if and only if $n = 0$.*

Proof. Enumerate the nonzero integers as

$$m_1, m_2, m_3, \dots$$

(for example $1, -1, 2, -2, 3, -3, \dots$). For each $j \geq 1$, choose a prime q_j such that $q_j \nmid m_j$; for instance, any prime $q_j > |m_j|$ will do.

We construct b_j inductively. Suppose $b_1, \dots, b_{j-1}$ have already been chosen. Since $q_j \nmid m_j$, one has

$$\gcd(-m_j, q_j^2) = 1.$$

By Dirichlet's theorem on primes in arithmetic progressions, the residue class

$$x \equiv -m_j \pmod{q_j^2}$$

contains infinitely many primes. Choose one such prime b_j so large that

$$b_j > \max\{b_{j-1},\, g(j),\, q_j^2 + |m_j|\}.$$

(For $j = 1$ the condition involving b_{j-1} is omitted.) This defines a strictly increasing sequence of primes with $b_j > g(j)$ for every j.

We now verify properties (ii) and (iii).

First, if $n = 0$, then $n + b_k = b_k$ is prime for every k by construction. Since every prime is squarefree, $n = 0$ also satisfies the squarefree condition.

Now let $n \neq 0$. Then $n = m_j$ for some index j. By construction,

$$b_j + n = b_j + m_j \equiv 0 \pmod{q_j^2}.$$

Moreover,

$$b_j + n \geq b_j - |m_j| > q_j^2,$$

so $b_j + n$ is a positive multiple of q_j^2 strictly larger than q_j^2. Therefore $b_j + n$ is composite, hence not prime. This proves that no nonzero shift can satisfy property (ii).

The same congruence also shows that

$$q_j^2 \mid (b_j + n),$$

so $b_j + n$ is not squarefree. Hence no nonzero shift can satisfy property (iii).

Thus $n = 0$ is the unique shift for which all $n + b_k$ are prime, and also the unique shift for which all $n + b_k$ are squarefree. □

The preceding theorem is stronger than what is asked in Problems (1) and (2), because it produces exactly one good shift rather than merely finitely many good shifts.

Corollary 2.2. *The answers to Problems* (1) *and* (2) *are negative. More precisely, given any function* $h\colon \mathbb{N} \to \mathbb{N}$ *with* $h(k) \to \infty$*, there exists a strictly increasing sequence of integers*

$$A = \{a_1 < a_2 < \cdots\}$$

with $a_k > h(k)$ *for every* k *such that there is exactly one shift* n *for which all* $n + a_k$ *are prime, and exactly one shift* n *for which all* $n + a_k$ *are squarefree.*

Proof. Apply Theorem 2.1 with a function g satisfying $g(k) \geq h(k) + N$ for some fixed positive integer N. Let $B = \{b_k\}$ be the resulting prime sequence, and define

$$a_k := b_k - N.$$

Then $a_k > h(k)$ for every k, and the sequence $A = \{a_k\}$ is strictly increasing and tends to ∞.

For any integer n we have

$$n + a_k = n + (b_k - N) = (n - N) + b_k.$$

Hence the shifts for A are exactly the shifts for B, translated by N. Since 0 is the unique good shift for B, the unique good shift for A is $n = N$. In particular there are not infinitely many such shifts. □

Remark 2.3. If one interprets Problems (1) and (2) as allowing all integer shifts, then Theorem 2.1 already answers them directly. Corollary 2.2 is included only to cover the common alternative convention that shifts must be positive.

3 The doubly exponential sequence $n + 2^{2^k}$ and Problem (3)

We now turn to the special sequence

$$U_k(n) := n + 2^{2^k} \qquad (k \geq 0).$$

The goal is to prove that no integer n makes every term $U_k(n)$ prime.

Theorem 3.1. *For every integer n, the sequence $\{U_k(n)\}_{k\geq 0}$ is* not *identically prime. Equivalently, there is no integer n such that*

$$n + 2^{2^k}$$

is prime for every $k \geq 0$.

The proof splits into three cases.

Case 1: n is even

If n is even, then $U_k(n)$ is even for every k. Since $2^{2^k} \to \infty$, for all sufficiently large k one has $U_k(n) > 2$, so $U_k(n)$ is composite. Thus an even n can never work.

Case 2: $n = 1$

In this case,

$$U_k(1) = 2^{2^k} + 1,$$

the kth Fermat number. It is enough to exhibit one composite term.

Lemma 3.2. *The number $2^{32} + 1$ is divisible by 641.*

Proof. We use the classical identities

$$641 = 5^4 + 2^4 = 5 \cdot 2^7 + 1.$$

The second identity implies

$$5 \cdot 2^7 \equiv -1 \pmod{641}.$$

Raising to the fourth power gives

$$5^4\, 2^{28} \equiv 1 \pmod{641}.$$

The first identity implies $5^4 \equiv -2^4 \pmod{641}$, so substituting into the previous congruence yields

$$-2^4\, 2^{28} \equiv 1 \pmod{641},$$

that is,

$$-2^{32} \equiv 1 \pmod{641}.$$

Therefore

$$2^{32} + 1 \equiv 0 \pmod{641},$$

so $641 \mid (2^{32} + 1)$. □

By Lemma 3.2, the term $U_5(1) = 2^{32} + 1$ is composite. Hence $n = 1$ does not work.

Case 3: n is odd and $n \neq 1$

This is the only genuinely nontrivial case. The argument is based on a recurrence of divisors.

Lemma 3.3. *Let n be an odd integer with $n \neq 1$, and put*

$$r := v_2(n-1).$$

Suppose that for some integer $t \geq r$ the number

$$p := U_t(n) = n + 2^{2^t}$$

is prime. Then there exists an integer $L \geq 1$ such that

$$p \mid U_{t+L}(n).$$

In fact,

$$p \mid U_{t+mL}(n) \qquad \text{for every } m \geq 1.$$

Proof. Since n is odd, the prime $p = n + 2^{2^t}$ is odd. Let

$$d := \mathrm{ord}_p(2),$$

the multiplicative order of 2 modulo p. By Fermat's little theorem, $2^{p-1} \equiv 1 \pmod{p}$, so $d \mid (p-1)$.

Now

$$p - 1 = (n-1) + 2^{2^t}.$$

Because $r = v_2(n-1)$ and $t \geq r$, we have

$$v_2\big(2^{2^t}\big) = 2^t > r.$$

Since the two summands $(n-1)$ and 2^{2^t} have different 2-adic valuations, the smaller valuation survives in the sum:

$$v_2(p-1) = v_2\big((n-1) + 2^{2^t}\big) = v_2(n-1) = r.$$

As $d \mid (p-1)$, it follows that

$$v_2(d) \le r \le t.$$

Consider the order of 2^{2^t} modulo p. A standard order formula gives

$$\operatorname{ord}_p(2^{2^t}) = \frac{d}{\gcd(d, 2^t)}.$$

Indeed, if an element has order d, then its mth power has order $d/\gcd(d,m)$, because one must solve the divisibility condition $d \mid ms$. Since $v_2(d) \le t$, the power 2^t contains the entire 2-part of d. Hence

$$M := \operatorname{ord}_p(2^{2^t}) = \frac{d}{\gcd(d, 2^t)}$$

is an odd positive integer.

Because M is odd, the residue class of 2 is invertible modulo M. Therefore some positive integer L satisfies

$$2^L \equiv 1 \pmod{M}.$$

(For example, when $M > 1$ one may take $L = \operatorname{ord}_M(2)$, and when $M = 1$ any $L \ge 1$ works.) Since 2^{2^t} has order M modulo p, exponents that are congruent modulo M give the same power. Thus

$$\left(2^{2^t}\right)^{2^L} \equiv \left(2^{2^t}\right)^1 \pmod{p}.$$

Equivalently,

$$2^{2^{t+L}} \equiv 2^{2^t} \pmod{p}.$$

Adding n to both sides gives

$$U_{t+L}(n) = n + 2^{2^{t+L}} \equiv n + 2^{2^t} = p \equiv 0 \pmod{p}.$$

So $p \mid U_{t+L}(n)$. Exactly the same argument works for every multiple mL, because $2^{mL} \equiv 1 \pmod{M}$ for all $m \ge 1$. Hence

$$p \mid U_{t+mL}(n) \qquad (m \ge 1).$$

□

We can now finish the odd case. Suppose, for contradiction, that $U_k(n)$ is prime for every $k \ge 0$. Since $r = v_2(n-1)$ is finite and every term is prime by assumption, the specific term

$$p := U_r(n) = n + 2^{2^r}$$

is prime. Applying Lemma 3.3 with $t = r$, we obtain an integer $L \ge 1$ such that

$$p \mid U_{r+L}(n).$$

But $r + L > r$, so

$$U_{r+L}(n) = n + 2^{2^{r+L}} > n + 2^{2^r} = p.$$

Therefore $U_{r+L}(n)$ is a multiple of the smaller positive integer p, and is therefore composite. This contradicts the assumption that every $U_k(n)$ is prime.

The three cases together prove Theorem 3.1.

4 Conclusion

The three problems listed in the introduction have the following answers.

(1) No. In fact, for any prescribed growth rate there exists a sequence $A = \{a_k\}$ for which there is exactly one shift n such that every $n + a_k$ is prime.

(2) No. The same construction shows that, for any prescribed growth rate, one can arrange that there is exactly one shift n such that every $n + a_k$ is squarefree.

(3) No. There is no integer n for which every number $n + 2^{2^k}$ is prime.